Algebra Tutorial

Homomorphisms and Isomorphisms

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Problems

Problem 1.

Determine whether each map is a group homomorphism. If it is, find its kernel and image.

(a)
$$f : \mathbb{Z} \to \mathbb{Z}, \qquad f(n) = 3n.$$

(b)
$$f : \mathbb{Z} \to \mathbb{Z}, \qquad f(n) = n + 1.$$

(c)
$$f : \mathbb{Z} \to \mathbb{Z}_6, \qquad f(n) = [2n]_6.$$

Problem 2.

Let
$$f : \mathbb{Z}_{12} \to \mathbb{Z}_6, \qquad f([x]_{12}) = [x]_6.$$

(a) Prove that f is a well-defined group homomorphism.

(b) Find $\ker(f)$.

(c) Find $\mathrm{im}(f)$.

(d) Is f injective? Is it surjective?

Problem 3.

Let
$$f : (\mathbb{R}, +) \to (\mathbb{R}_{>0}, \cdot), \qquad f(x) = e^x.$$

(a) Prove that f is a group homomorphism.

(b) Determine $\ker(f)$.

(c) Determine $\mathrm{im}(f)$.

(d) Conclude that f is an isomorphism.

Problem 4.

Let

$$f : \mathbb{Z}_{20} \to \mathbb{Z}_{20}, \qquad f([x]_{20}) = [6x]_{20}.$$

Determine whether f is a group homomorphism. Find its kernel and image, and determine whether f is injective or surjective.

Repeat the problem with

$$f([x]_{20}) = [7x]_{20}.$$

Problem 5.

Let

$$f : G \to H$$

be a group homomorphism.

Prove the following statements:

(a) $f(e_G) = e_H$.

(b) $f(g^{-1}) = f(g)^{-1}$ for every $g \in G$.

(c) If f is injective, then $\ker(f) = \{e_G\}$.

Problem 6.

Let

$$G = \mathbb{Z}_4 \times \mathbb{Z}_2$$

and

$$H = \mathbb{Z}_8.$$

Determine whether G and H are isomorphic as groups.

Give a proof using an invariant of groups. In particular, compare the orders of elements in the two groups.

Problem 7.

Define

$$f : \mathbb{Z}_6 \times \mathbb{Z}_4 \longrightarrow \mathbb{Z}_{12}$$

by

$$f([a]_6, [b]_4) = [2a + 3b]_{12}.$$

(a) Prove that f is a well-defined group homomorphism.

(b) Determine $\ker(f)$.

(c) Determine $\mathrm{im}(f)$.

(d) Is f injective? Is it surjective?

Problem 8.

Let G and H be finite groups and let

$$f : G \to H$$

be a group homomorphism.

Suppose that

$$|G| = |H|.$$

Prove that the following statements are equivalent:

(a) f is injective;

(b) f is surjective;

(c) f is an isomorphism.

Problem 9.

Let

$$G = \mathbb{Z}_{12}$$

and consider the subgroup

$$H = \langle [4]_{12} \rangle.$$

(a) Determine H.

(b) Define

$$f : \mathbb{Z}_{12} \to \mathbb{Z}_3, \qquad f([x]_{12}) = [x]_3.$$

Find $\ker(f)$.

(c) Show that

$$\mathbb{Z}_{12}/H \cong \mathbb{Z}_3.$$

Problem 10. Challenge

Let G be a group and let

$$\varphi : G \to G$$

be a group homomorphism satisfying

$$\varphi(\varphi(g)) = g \qquad \text{for every } g \in G.$$

(a) Prove that φ is an isomorphism.

(b) Prove that

$$G = \ker(\varphi) \times \mathrm{im}(\varphi)$$

(c) Give an example of such a homomorphism on a familiar group.